Theorem 9. *The f-divergences are invariant by diffeomorphisms $m(x)$ of the sample space $\mathcal{X}$: Let $Y = m(X)$, and $X_i \sim p_i$ with $Y_i = m(X_i) \sim q_i$. Then we have $I_f[q_1 : q_2] = I_f[p_1 : p_2]$.*

Example 4. *Consider the exponential distributions and the Rayleigh distributions which are related by:*

$$X \sim \text{Exponential}(\lambda) \Leftrightarrow Y = m(X) = \sqrt{X} \sim \text{Rayleigh}\left(\sigma = \frac{1}{\sqrt{2\lambda}}\right).$$

The densities of the exponential distributions are defined by

$$p_\lambda(x) = \lambda \exp(-\lambda x) \text{ with support } \mathcal{X} = [0, \infty),$$

and the densities of the Rayleigh distributions are defined by

$$q_\sigma(x) = \frac{x}{\sigma^2} \exp\left(-\frac{x^2}{2\sigma^2}\right) \text{ with support } \mathcal{X} = [0, \infty).$$

We have

$$D_{\mathrm{KL}}\left[q_{\sigma_1} : q_{\sigma_2}\right] = \log\left(\frac{\lambda_2^2}{\lambda_1^2}\right) + \frac{\sigma_1^2 - \sigma_2^2}{\sigma_2^2}.$$

It follows that

$$\begin{aligned} D_{\mathrm{KL}}\left[p_{\lambda_1} : p_{\lambda_2}\right] &= D_{\mathrm{KL}}\left[q\frac{1}{\sqrt{2\lambda_1}} : q\frac{1}{\sqrt{2\lambda_1}}\right] \\ &= \log\frac{2\lambda_1}{2\lambda_2} + 2\lambda_2\left(\frac{1}{2\lambda_1} - \frac{1}{\lambda_2}\right) \\ &= \log\left(\frac{\lambda_1}{\lambda_2}\right) + \frac{\lambda_2}{\lambda_1} - 1. \end{aligned}$$

A remarkable property is that invariant standard f-divergences yield the Fisher information matrix and the α-connections. Indeed, the invariant standard f-divergences is related infinitesimally to the Fisher metric as follows:

$$I_f[p(x;\theta) : p(x;\theta + \mathrm{d}\theta)] = \int p(x;\theta) f\left(\frac{p(x;\theta + \mathrm{d}\theta)}{p(x;\theta)}\right) \mathrm{d}\mu(x) \tag{116}$$

$$\overset{\Sigma}{=} \frac{1}{2} {}_F g_{ij}(\theta) \mathrm{d}\theta^i \mathrm{d}\theta^j \tag{117}$$

A *statistical parameter divergence* D on a parametric family of distributions $\mathcal{P}$ yields an equivalent *parameter divergence* ${}_{\mathcal{P}}D$:

$${}_{\mathcal{P}}D(\theta : \theta') := D[p(x;\theta) : p(x;\theta')]. \tag{118}$$

Thus we can build the information manifold induced by this parameter divergence ${}_{\mathcal{P}}D(\cdot : \cdot)$. For ${}_{\mathcal{P}}D(\cdot : \cdot) = I_f[\cdot : \cdot]$, the induced ± 1-divergence connections ${}^{I_f}_{\mathcal{P}}\nabla := {}^{\mathcal{P}I_f}\nabla$ and ${}^{(I_f)^*}_{\mathcal{P}}\nabla := {}^{\mathcal{P}I_f^*}\nabla$ are precisely the *expected* $\pm\alpha$*-connections* (derived from the exponential/mixture connections) with:

$$\alpha = 2f'''(1) + 3. \tag{119}$$

Thus the invariant connections which coincide with the connections induced by the invariant statistical divergences are the expected α-connections. Note that the curvature of an expected α-connection depends both on α and on the considered statistical model [64].

3.11 Fisher-Rao expected Riemannian manifolds: $(\mathcal{P}, {}_{\mathcal{P}}g)$

Historically, a first manifold modeling of a regular parametric family of distributions $\mathcal{P} = \{p_\theta(x)\}_\theta$ was to consider the *Fisher Information Matrix* (FIM) as the Riemannian metric tensor g (see [53, 111]), with:

$$ {}_{\mathcal{P}}I(\theta) := E_{p_\theta}\left[\partial_i l \partial_j l\right], \tag{120}$$

where $\partial_i l := \frac{\partial}{\partial \theta_i} \log p(x;\theta)$. Under some regularity conditions, we can rewrite the FIM:

$$ {}_{\mathcal{P}}I(\theta) := -E_{p_\theta}\left[\partial_i \partial_j l\right]. \tag{121}$$

The Riemannian geodesic metric distance D_ρ is commonly called the *Fisher-Rao distance*:

$$D_\rho(p_{\theta_1}, p_{\theta_2}) = \int_0^1 \sqrt{\dot{\gamma}(t)^\top g_{\gamma(t)} \dot{\gamma}(t)} \mathrm{d}t, \tag{122}$$

where γ denotes the geodesic passing through $\gamma(0) = \theta_1$ and $\gamma(1) = \theta_2$. The Fisher-Rao distance can also be defined as the shortest path length: $D_\rho(p_{\theta_1}, p_{\theta_2}) = \inf_\gamma \int_0^1 \sqrt{\dot{\gamma}(t)^\top g_{\gamma(t)} \dot{\gamma}(t)} \mathrm{d}t$.

Definition 6 (Fisher-Rao distance)**.** *The Fisher-Rao distance is the geodesic metric distance of the Fisher-Riemannian manifold* $(\mathcal{P}, {}_{\mathcal{P}}g)$.

Let us give some examples of Fisher-Riemannian manifolds:

- The Fisher-Riemannian manifold of the family of categorical distributions (also called finite discrete distributions in [8]) amount to the spherical geometry [58] (spherical manifold).
- The Fisher-Riemannian manifold of the family of bivariate location-scale families amount to hyperbolic geometry (hyperbolic manifold).
- The Fisher-Riemannian manifold of the family of location families amount to Euclidean geometry (Euclidean manifold).

The first fundamental form of the Riemannian geometry is $\mathrm{d}s^2 = \langle \mathrm{d}x, \mathrm{d}x \rangle \overset{\Sigma}{=} g_{ij} \mathrm{d}x^i \mathrm{d}x^j$ where $\mathrm{d}s$ denotes the line element.

This Riemannian geometric structure applied to a family of parametric probability distributions was first proposed by Harold Hotelling [53] (in a handwritten note of 1929, reprinted typeset in [123]) and independently later by C. R. Rao [111] (1945, reprinted in [110]). In a similar vein, Jeffreys [55] proposed to use the volume element of a manifold as an invariant prior: The eponym Jeffreys prior in 1946.

Notice that for a parametric family of probability distributions $\mathcal{P}$, the Riemannian structure $(\mathcal{P}, {}_{\mathcal{P}}g)$ coincides with the self-dual conjugate connection manifold $(\mathcal{P}, {}_{\mathcal{P}}g, {}^{I_f}_{\mathcal{P}}\nabla, {}^{I_f}_{\mathcal{P}}\nabla^*)$ induced by a *symmetric* f-divergence like the squared Hellinger divergence.

The exponential map $\exp_p$ at point $p \in M$ provides a way to map back a vector $v \in T_p$ to a point $\exp_p(v) \in M$ (when well-defined). The exponential map can be used to parameterize a geodesic γ with $\gamma(0) = p$ and unit tangent vector $\dot{\gamma}(0) = v$: $t \mapsto \exp_p(tv)$. For geodesically complete manifolds, the exponential map is defined everywhere.

3.12 The monotone α-embeddings and the metric gauge freedom

Another common mathematically equivalent expression of the FIM [25] is given by:

$$I_{ij}(\theta) := 4 \int \partial_i \sqrt{p(x;\theta)} \partial_j \sqrt{p(x;\theta)} \mathrm{d}\mu(x). \tag{123}$$

This form of the FIM is well-suited to prove that the FIM is always a positive semi-definite matrix [25] ($I(\theta) \succeq 0$). It turns out that we can define a family of *equivalent representations* of the FIM using the *α-embedding* [139] of the parametric family.

First, we define the *α-representation* of densities $l^\alpha(x;\theta) := k_\alpha(p(x;\theta))$ with:

$$k_\alpha(u) := \begin{cases} \frac{2}{1-\alpha} u^{\frac{1-\alpha}{2}}, & \text{if } \alpha \neq 1, \\ \log u, & \text{if } \alpha = 1. \end{cases} \tag{124}$$

The function $l^\alpha(x;\theta)$ is called the *α-likelihood* function. Then the α-representation of the FIM, the α-FIM for short, is expressed as:

$$I^\alpha_{ij}(\theta) := \int \partial_i l^\alpha(x;\theta) \partial_j l^{-\alpha}(x;\theta) \mathrm{d}\mu(x). \tag{125}$$

We can rewrite compactly the α-FIM, as $I^\alpha_{ij}(\theta) = \int \partial_i l^\alpha \partial_j l^{-\alpha} \mathrm{d}\mu(x)$. Expanding the α-FIM, we get:

$$I^\alpha_{ij}(\theta) = \begin{cases} \frac{1}{1-\alpha^2} \int \partial_i p(x;\theta)^{\frac{1-\alpha}{2}} \partial_j p(x;\theta)^{\frac{1+\alpha}{2}} \mathrm{d}\mu(x) & \text{for } \alpha \neq \pm 1 \\ \int \partial_i \log p(x;\theta) \partial_j p(x;\theta) \mathrm{d}\mu(x) & \text{for } \alpha \in \{-1, 1\} \end{cases} \tag{126}$$

The 1-representation of the density is called the *logarithmic representation* (or e-representation), the -1-representation the *mixture representation* (or m-representation), and its 0-representation is called the *square root representation*. The set of *α-scores* vectors $B_\alpha := \{\partial_i l^\alpha\}_i$ are interpreted as the tangent basis vectors of the α-base B_α. Thus the FIM is α-independent.

Furthermore, the α-representation of the FIM can be rewritten under mild conditions [25] as:

$$I^\alpha_{ij}(\theta) = -\frac{2}{1+\alpha} \int p(x;\theta)^{\frac{1+\alpha}{2}} \partial_i \partial_j l^\alpha(x;\theta) \mathrm{d}\mu(x). \tag{127}$$

Since we have:

$$\partial_i \partial_j l^\alpha(x;\theta) = p^{\frac{1-\alpha}{2}} \left(\partial_i \partial_j l + \frac{1-\alpha}{2} \partial_i l \partial_j l \right), \tag{128}$$

it follows that:

$$I^\alpha_{ij}(\theta) = -\frac{2}{1+\alpha} \left(-I_{ij}(\theta) + \frac{1-\alpha}{2} I_{ij} \right) = I_{ij}(\theta). \tag{129}$$

Notice that when $\alpha = 1$, we recover the equivalent expression of the FIM (under mild conditions):

$$I^1_{ij}(\theta) = -E[\nabla^2 \log p(x;\theta)]. \tag{130}$$

In particular, when the family is an exponential family [84] with cumulant function $F(\theta)$ (satisfying the mild conditions), we have:

$$I(\theta) = \nabla^2 F(\theta). \tag{131}$$

Zhang [139, 69] further discussed the representation/reference biduality which was confounded in the α-geometry.

Gauge freedom of the Riemannian metric tensor has been investigated under the framework of (ρ, τ)-monotone embeddings [139, 98, 69] in information geometry: Let ρ and τ be two strictly increasing functions, and f a strictly convex function such that $f'(\rho(u)) = \tau(u)$ (with f^* denoting its convex conjugate). Observe that the set of strictly increasing real-valued univariate functions has a group structure for the group operation chosen as the functional composition $\circ$. Let us write $p_\theta(x) = p(x;\theta)$.

The (ρ, τ)-metric tensor ${}^{\rho,\tau}g(\theta) = [\,{}^{\rho,\tau}g_{ij}(\theta)]_{ij}$ can be derived from the (ρ, τ)-divergence:

$$D_{\rho,\tau}(p:q) = \int \left(f(\rho(p(x))) + f^*(\tau(q(x))) - \rho(p(x))\tau(q(x)) \right) \mathrm{d}\nu(x) \tag{132}$$